

MAT 022A Final Exam

September 13, 2019

Name: Final Exam Key

SID: _____

Instructions: Do not turn the page until instructed to do so. No external help or calculators allowed. Please show your work (with the exception of row operations) and explain your answers if necessary. Box your answer if applicable.

1. 5 years ago, Aaron was 3 times as old as Beth. Today, Aaron is twenty years older than Beth. How old are Aaron and Beth?

$$a - 5 = 3(b - 5)$$

$$a = b + 20$$

$$a - 5 = 3b - 15$$

$$a = 3b - 10$$

$$b + 20 = 3b - 10$$

$$30 = 2b$$

$$b = 15$$

$$a = 15 + 20$$

$$a = 35$$

Aaron is 35 years old.

Beth is 15 years old.

2. Find the determinants of the following matrices given that

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & -2 & 2 & 2 & 2 \\ 3 & -3 & -3 & 3 & 3 \\ 4 & -4 & -4 & -4 & 4 \\ 5 & -5 & -5 & -5 & -5 \end{bmatrix} \text{ and } \det(A) = 1920.$$

$$\text{a. } A_1 = \begin{bmatrix} 5 & -5 & -5 & -5 & -5 \\ 2 & -2 & 2 & 2 & 2 \\ 3 & -3 & -3 & 3 & 3 \\ 4 & -4 & -4 & -4 & 4 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} \det(A_1) &= \det(P_{15}A) \\ &= \det(P_{15})\det(A) \\ &= -1 \cdot 1920 \\ &= -1920 \end{aligned}$$

$$\text{b. } A_2 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 20 & -20 & 20 & 20 & 20 \\ 3 & -3 & -3 & 3 & 3 \\ 4 & -4 & -4 & -4 & 4 \\ 5 & -5 & -5 & -5 & -5 \end{bmatrix}$$

$$\begin{aligned} \det(A_2) &= \det(R_2(10)A) \\ &= \det(R_2(10))\det(A) \\ &= 10 \cdot 1920 \\ &= 19200 \end{aligned}$$

$$\text{c. } A_3 = \begin{bmatrix} 10 & 10 & 10 & 10 & 10 \\ 4 & -4 & -4 & -4 & 4 \\ 2 & -2 & 2 & 2 & 2 \\ 3 & -3 & -3 & 3 & 3 \\ 50 & -50 & -50 & -50 & -50 \end{bmatrix}$$

$$\begin{aligned} \det(A_3) &= \det(P_{24}P_{34}R_1(10)R_5(10)A) \\ &= \det(P_{24})\det(R_1(10))\det(R_5(10))\det(A) \\ &= (-1)(-1)(10)(10)(1920) \\ &= 192000 \end{aligned}$$

3. Find the eigenvalues and the associated eigenvectors of $A = \begin{bmatrix} 3 & 2 & 2 \\ 0 & 4 & 3 \\ 0 & 5 & 2 \end{bmatrix}$.

$$\begin{aligned} (3-\lambda) \begin{vmatrix} 4-\lambda & 3 \\ 5 & 2-\lambda \end{vmatrix} &= (3-\lambda) [(4-\lambda)(2-\lambda) - 15] \\ &= (3-\lambda)(8 - 6\lambda + \lambda^2 - 15) \\ &= (3-\lambda)(\lambda^2 - 6\lambda - 7) \\ &= (3-\lambda)(\lambda-7)(\lambda+1) \end{aligned}$$

$$\lambda = -1, 3, 7$$

$$\begin{aligned} \underline{\lambda_1 = -1}: \begin{bmatrix} 4 & 2 & 2 \\ 0 & 5 & 3 \\ 0 & 5 & 3 \end{bmatrix} &\sim \begin{bmatrix} 4 & 2 & 2 \\ 0 & 5 & 3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{3}{5} \\ 0 & 0 & 0 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & \frac{1}{5} \\ 0 & 1 & \frac{3}{5} \\ 0 & 0 & 0 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} -1 \\ -3 \\ 5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \underline{\lambda_2 = 3}: \begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 3 \\ 0 & 5 & -1 \end{bmatrix} &\sim \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -6 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\ \vec{v}_2 &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \underline{\lambda_3 = 7}: \begin{bmatrix} -4 & 2 & 2 \\ 0 & -3 & 3 \\ 0 & 5 & -5 \end{bmatrix} &\sim \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \\ \vec{v}_3 &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

4. Find the QR-decomposition of $A = \begin{bmatrix} 1 & 2 & -3 \\ -1 & -1 & 1 \\ 1 & 3 & 1 \end{bmatrix}$.

$$\vec{v}_1^\perp = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad v_2^\perp = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} - \frac{2+1+3}{1+1+1} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} - \frac{6}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \vec{v}_3^\perp &= \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} - \frac{-3-1+1}{1+1+1} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} - \frac{0+1+1}{0+1+1} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} - \frac{-3}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 & -2 \\ -1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 2\sqrt{3} & -\sqrt{3} \\ 0 & \sqrt{2} & \sqrt{2} \\ 0 & 0 & \sqrt{6} \end{bmatrix}$$

5. Find the line $y = \beta_0 + \beta_1 x$ that best fits the points $(4, 1), (6, 6), (8, 5)$.

$$\tilde{X} = \begin{bmatrix} 1 & 4 \\ 1 & 6 \\ 1 & 8 \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 1 \\ 6 \\ 5 \end{bmatrix} \quad \begin{array}{r} 16 \\ 36 \\ + 64 \\ \hline 116 \end{array}$$

$$\tilde{X}^T \tilde{X} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 6 & 8 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 6 \\ 1 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 18 \\ 18 & 116 \end{bmatrix}$$

$$\tilde{X}^T \vec{y} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 6 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 80 \end{bmatrix} \quad \begin{array}{r} 4 \\ 18 \\ \times 6 \\ \hline 108 \end{array}$$

$$\left[\begin{array}{cc|c} 3 & 18 & 12 \\ 18 & 116 & 80 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 6 & 4 \\ 0 & 8 & 8 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 6 & 4 \\ 0 & 1 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 1 \end{array} \right] \quad \begin{array}{l} \beta_0 = -2 \\ \beta_1 = 1 \end{array}$$

$$y = -2 + x$$

6. Find the singular value decomposition of $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$.

$$A^T A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ -2 & 8 \end{bmatrix}$$

$$(5-\lambda)(8-\lambda)-4=0$$

$$\lambda^2 - 13\lambda + 36 = 0$$

$$(\lambda-4)(\lambda-9)=0$$

$$\lambda = 4, 9$$

$$\sigma = 2, 3$$

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\underline{\lambda=4}: \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\underline{\lambda=9}: \begin{bmatrix} -4 & -2 \\ -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$\vec{u}_1 = A\vec{v}_1 = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \sim \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\vec{u}_2 = A\vec{v}_2 = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \end{bmatrix} \sim \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix}$$

$$A = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

7. Show that if A is a nonsingular symmetric matrix, then A^{-1} is also symmetric.

$$\begin{aligned}(A^{-1})^T &= (A^T)^{-1} \\ &= (A)^{-1} \\ &= A^{-1}\end{aligned}$$

Therefore A^{-1} is symmetric.

8. Let A and B be $n \times m$ matrices such that $\text{rref}(A) = \text{rref}(B)$. Show that if $\vec{x}$ is in the null space of A , then $\vec{x}$ is in the null space of B .

$$A\vec{x} = \vec{0}$$

$$\text{rref}(A)\vec{x} = \vec{0}$$

$$\text{rref}(B)\vec{x} = \vec{0}$$

$$B\vec{x} = \vec{0}$$

Therefore $\vec{x}$ is in the null space of B .

9. Let λ be an eigenvalue of A with eigenvector $\vec{x}$. Show that λ^3 is an eigenvalue of A^3 with eigenvector $\vec{x}$.

$$A\vec{x} = \lambda\vec{x}$$

$$A^2\vec{x} = \lambda A\vec{x}$$

$$= \lambda \cdot \lambda\vec{x}$$

$$= \lambda^2\vec{x}$$

$$A^3\vec{x} = \lambda^2 A\vec{x}$$

$$= \lambda^2 \cdot \lambda\vec{x}$$

$$A^3\vec{x} = \lambda^3\vec{x}$$

Therefore λ^3 is an eigenvalue of A^3 with eigenvector $\vec{x}$.

10. Show that if P is an orthogonal matrix, then $\det(P) = \pm 1$.

$$P^T = P^{-1}$$

$$\det(P^T) = \det(P^{-1})$$

$$\det(P) = \frac{1}{\det(P)}$$

$$\det(P) \det(P) = 1$$

$$[\det(P)]^2 = 1$$

$$\det(P) = \pm 1$$